

Bayescoverage APP - overview

Leontine Alkema, May 11 2026

Installation info is given in the readme.

Just loading all packages here to avoid issues (we may be able to remove some, eg brms).

```
library(bayescoveragemodel)
library(localhierarchy) # for check_nas...
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
#library(ggnewscale)
#library(ggpubr)
library(haven)
#library(stringr)
#library(readxl)
library(brms) # brms required for routine stuff
```

Loading required package: Rcpp

Loading 'brms' package (version 2.23.0). Useful instructions can be found by typing `help('brms')`. A more detailed introduction to the package is available through `vignette('brms_overview')`.

Attaching package: 'brms'

The following object is masked from 'package:stats':

ar

```
options(cmdstanr_warn_inits = FALSE)
```

Where data are stored:

```
data_folder <- "data_raw" # for survey data
routinedata_dir <- here::here("private/routinedata/")
```

Choose an indicator. Naming comes from survey data, options include

- c("anc1trimester", "vmsl", "ideliv", "vdpt", "anc4") , with routine as available
- c("bfexcl0_5", "ancq8", "cci", "sba"), no routine data

```
indicator_select <- "anc4" #"cci"#"vdpt"#"anc4"
```

Read in survey data and process

```
dat0 <- read_dta(here::here(data_folder, "ICEH_all.long.dta")) %>%
  rename(r = r_raw, se = se_raw)
# process for indicator_select
dat <- process_data(
  dat = dat0,
  regions_dat = bayescoveragemodel::regions_all,
  indicator = indicator_select)
```

```
[1] "For ANC4, we combined anc42 and anc43 and call it anc4."
[1] "Some clusters are NA, these observations are excluded for now."
[1] "Observations with missing clusters"
# A tibble: 6 x 3
  iso    year cluster
<chr> <dbl> <chr>
1 XKX    2013 <NA>
2 XKX    2019 <NA>
3 PBA    2010 <NA>
```

```

4 PPU      2011 <NA>
5 PIL      2006 <NA>
6 PIL      2011 <NA>
[1] "Minor TO DO: Figure out exact reference years, ie different reference period for anc4."
[1] "Some SEs are NA, these observations are excluded for now."
[1] "TO DO: add imputation for missing SEs."
[1] "We add record_id_fixed to the data."
[1] "Currently no biases assigned."
[1] "Column possible_outlier not included in data, no pre-defined outliers used"
[1] "Remove data that's not DHS or MICS, number of obs removed:"
[1] 6

```

Do we want to add routine data?

```

add_routine <- ifelse(
  indicator_select %in% c("anc1trimester", "vmsl", "ideliv", "vdpt", "anc4"),
  TRUE,
  FALSE)
add_routine

```

```
[1] TRUE
```

If yes, read in routine data. Some processing, we'll move this processing into a function.

```

if (!add_routine){
  routine_dat <- routine_dat_use <- NULL
} else {
  routine_dat0 <- read_csv(
    paste0(routinedata_dir, "/dat_combi_",
           indicator_select, ".csv")) %>%
    mutate(indicator = indicator_select)
  # routine_dat has indicator_name (name in routine) and
  # indicator (indicator name from survey data)
  routine_dat <-
    routine_dat0 %>%
    filter(worst_notmissing >= 90,
           worst_notoutliers >= 90,
           worst_abs_diff_level < 0.2,
           worst_rr >= 60) %>%
    # %>%
    # filter(!is.na(diff_roc))

```

```

# remove countries w/o any nonNA diff_roc
group_by(iso) %>%
filter(sum(!is.na(diff_roc)) > 0) %>%
ungroup() %>%
mutate(sd_routine_roc = 0,
       sd_routine = 0) %>%
mutate(worst_combi =
       worst_rr/100*worst_notmissing/100*worst_notoutliers)
}

```

Rows: 149 Columns: 27

-- Column specification -----

Delimiter: ","

chr (6): country, iso, indicator_name, country_denom_type, subcluster, cluster

dbl (20): countrycode, year, routine_value, median_estimate, max_survey_yr, ...

lgl (1): within_survey_period

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Fitting for one country, use ISO to select

```

# countries with survey data
# unique(dat$iso)
iso_select <- "RWA"
dat_use <-
  dat %>%
  filter(iso %in% iso_select)
if (add_routine){
  routine_dat_use <-
    routine_dat %>%
    filter(iso %in% iso_select)
}

```

```

fit <- fit_model(runstep = "local_national",
                 y = "invprobit_indicator",
                 se = "se_invprobit_indicator",
                 survey_df = dat_use,
                 routine_data = routine_dat_use,
                 #stan_file_path = stan_file_path_update,
                 chains = 4,

```

```

iter_sampling = 300,
iter_warmup = 150,
# validation_cutoff_year = 2010,
get_posteriors = TRUE)

```

```

[1] "We use a global fit, and take selected settings from there."
[1] "settings for the spline_degree and num_knots taken from global run"
[1] "settings for Ptilde_low taken from global run"
[1] "Setting for tstar taken from global run"
[1] "Settings for hierarchical settings taken from global run"
[1] "For hierarchical terms, we fix things up to the 2nd-lowest level."
[1] "We fix all sigmas of hierarchical models."
[1] "We take the data model setting from global fit."
[1] "We add smoothing"
[1] "We fix smoothing and data model parameters."
[1] "We filter data to be inside estimation period"
[1] "We use national data"
[1] "NOT using or producing aggregates"

```

Joining with `by = join\_by(iso)`

Running MCMC with 4 parallel chains...

Chain 1 Iteration: 1 / 450 [0%] (Warmup)

Chain 1 Iteration: 10 / 450 [2%] (Warmup)

Chain 1 Informational Message: The current Metropolis proposal is about to be rejected because of the following reasons:

Chain 1 Exception: normal_lpdf: Location parameter is nan, but must be finite! (in '/var/folders/c0a45c927d84.stan', line 401, column 10 to column 47)

Chain 1 If this warning occurs sporadically, such as for highly constrained variable types like real, it is probably safe to ignore. Chain 1 but if this warning occurs often then your model may be either severely ill-conditioned or misspecified.

Chain 1

Chain 2 Iteration: 1 / 450 [0%] (Warmup)

Chain 2 Iteration: 10 / 450 [2%] (Warmup)

Chain 2 Iteration: 20 / 450 [4%] (Warmup)

Chain 3 Iteration: 1 / 450 [0%] (Warmup)

Chain 3 Iteration: 10 / 450 [2%] (Warmup)

Chain 3 Iteration: 20 / 450 [4%] (Warmup)

```
Chain 3 Iteration: 30 / 450 [ 6%] (Warmup)
Chain 4 Iteration: 1 / 450 [ 0%] (Warmup)
Chain 4 Iteration: 10 / 450 [ 2%] (Warmup)
Chain 4 Iteration: 20 / 450 [ 4%] (Warmup)
Chain 4 Iteration: 30 / 450 [ 6%] (Warmup)
```

```
Chain 4 Informational Message: The current Metropolis proposal is about to be rejected because of the following reasons:
Chain 4 Exception: normal_lpdf: Location parameter is nan, but must be finite! (in '/var/folders/0a/0a45c927d84.stan', line 401, column 10 to column 47)
```

```
Chain 4 If this warning occurs sporadically, such as for highly constrained variable types like real, it is usually harmless.
Chain 4 but if this warning occurs often then your model may be either severely ill-conditioned or misspecified.
```

```
Chain 4
```

```
Chain 1 Iteration: 20 / 450 [ 4%] (Warmup)
Chain 1 Iteration: 30 / 450 [ 6%] (Warmup)
Chain 1 Iteration: 40 / 450 [ 8%] (Warmup)
Chain 2 Iteration: 30 / 450 [ 6%] (Warmup)
Chain 2 Iteration: 40 / 450 [ 8%] (Warmup)
Chain 2 Iteration: 50 / 450 [ 11%] (Warmup)
Chain 3 Iteration: 40 / 450 [ 8%] (Warmup)
Chain 3 Iteration: 50 / 450 [ 11%] (Warmup)
Chain 4 Iteration: 40 / 450 [ 8%] (Warmup)
Chain 4 Iteration: 50 / 450 [ 11%] (Warmup)
Chain 1 Iteration: 50 / 450 [ 11%] (Warmup)
Chain 1 Iteration: 60 / 450 [ 13%] (Warmup)
Chain 1 Iteration: 70 / 450 [ 15%] (Warmup)
Chain 2 Iteration: 60 / 450 [ 13%] (Warmup)
Chain 2 Iteration: 70 / 450 [ 15%] (Warmup)
Chain 2 Iteration: 80 / 450 [ 17%] (Warmup)
Chain 3 Iteration: 60 / 450 [ 13%] (Warmup)
Chain 3 Iteration: 70 / 450 [ 15%] (Warmup)
Chain 3 Iteration: 80 / 450 [ 17%] (Warmup)
Chain 4 Iteration: 60 / 450 [ 13%] (Warmup)
Chain 4 Iteration: 70 / 450 [ 15%] (Warmup)
Chain 1 Iteration: 80 / 450 [ 17%] (Warmup)
Chain 1 Iteration: 90 / 450 [ 20%] (Warmup)
Chain 1 Iteration: 100 / 450 [ 22%] (Warmup)
Chain 2 Iteration: 90 / 450 [ 20%] (Warmup)
Chain 2 Iteration: 100 / 450 [ 22%] (Warmup)
Chain 2 Iteration: 110 / 450 [ 24%] (Warmup)
Chain 2 Iteration: 120 / 450 [ 26%] (Warmup)
```

Chain 2 Iteration: 130 / 450 [28%] (Warmup)
Chain 3 Iteration: 90 / 450 [20%] (Warmup)
Chain 3 Iteration: 100 / 450 [22%] (Warmup)
Chain 3 Iteration: 110 / 450 [24%] (Warmup)
Chain 3 Iteration: 120 / 450 [26%] (Warmup)
Chain 4 Iteration: 80 / 450 [17%] (Warmup)
Chain 4 Iteration: 90 / 450 [20%] (Warmup)
Chain 4 Iteration: 100 / 450 [22%] (Warmup)
Chain 1 Iteration: 110 / 450 [24%] (Warmup)
Chain 1 Iteration: 120 / 450 [26%] (Warmup)
Chain 1 Iteration: 130 / 450 [28%] (Warmup)
Chain 1 Iteration: 140 / 450 [31%] (Warmup)
Chain 1 Iteration: 150 / 450 [33%] (Warmup)
Chain 1 Iteration: 151 / 450 [33%] (Sampling)
Chain 1 Iteration: 160 / 450 [35%] (Sampling)
Chain 1 Iteration: 170 / 450 [37%] (Sampling)
Chain 2 Iteration: 140 / 450 [31%] (Warmup)
Chain 2 Iteration: 150 / 450 [33%] (Warmup)
Chain 2 Iteration: 151 / 450 [33%] (Sampling)
Chain 2 Iteration: 160 / 450 [35%] (Sampling)
Chain 2 Iteration: 170 / 450 [37%] (Sampling)
Chain 2 Iteration: 180 / 450 [40%] (Sampling)
Chain 2 Iteration: 190 / 450 [42%] (Sampling)
Chain 2 Iteration: 200 / 450 [44%] (Sampling)
Chain 3 Iteration: 130 / 450 [28%] (Warmup)
Chain 3 Iteration: 140 / 450 [31%] (Warmup)
Chain 3 Iteration: 150 / 450 [33%] (Warmup)
Chain 3 Iteration: 151 / 450 [33%] (Sampling)
Chain 3 Iteration: 160 / 450 [35%] (Sampling)
Chain 3 Iteration: 170 / 450 [37%] (Sampling)
Chain 3 Iteration: 180 / 450 [40%] (Sampling)
Chain 4 Iteration: 110 / 450 [24%] (Warmup)
Chain 4 Iteration: 120 / 450 [26%] (Warmup)
Chain 4 Iteration: 130 / 450 [28%] (Warmup)
Chain 4 Iteration: 140 / 450 [31%] (Warmup)
Chain 4 Iteration: 150 / 450 [33%] (Warmup)
Chain 4 Iteration: 151 / 450 [33%] (Sampling)
Chain 4 Iteration: 160 / 450 [35%] (Sampling)
Chain 4 Iteration: 170 / 450 [37%] (Sampling)
Chain 1 Iteration: 180 / 450 [40%] (Sampling)
Chain 1 Iteration: 190 / 450 [42%] (Sampling)
Chain 1 Iteration: 200 / 450 [44%] (Sampling)
Chain 1 Iteration: 210 / 450 [46%] (Sampling)

Chain 1 Iteration: 220 / 450 [48%] (Sampling)
Chain 1 Iteration: 230 / 450 [51%] (Sampling)
Chain 1 Iteration: 240 / 450 [53%] (Sampling)
Chain 1 Iteration: 250 / 450 [55%] (Sampling)
Chain 2 Iteration: 210 / 450 [46%] (Sampling)
Chain 2 Iteration: 220 / 450 [48%] (Sampling)
Chain 2 Iteration: 230 / 450 [51%] (Sampling)
Chain 2 Iteration: 240 / 450 [53%] (Sampling)
Chain 2 Iteration: 250 / 450 [55%] (Sampling)
Chain 2 Iteration: 260 / 450 [57%] (Sampling)
Chain 2 Iteration: 270 / 450 [60%] (Sampling)
Chain 2 Iteration: 280 / 450 [62%] (Sampling)
Chain 3 Iteration: 190 / 450 [42%] (Sampling)
Chain 3 Iteration: 200 / 450 [44%] (Sampling)
Chain 3 Iteration: 210 / 450 [46%] (Sampling)
Chain 3 Iteration: 220 / 450 [48%] (Sampling)
Chain 3 Iteration: 230 / 450 [51%] (Sampling)
Chain 3 Iteration: 240 / 450 [53%] (Sampling)
Chain 3 Iteration: 250 / 450 [55%] (Sampling)
Chain 4 Iteration: 180 / 450 [40%] (Sampling)
Chain 4 Iteration: 190 / 450 [42%] (Sampling)
Chain 4 Iteration: 200 / 450 [44%] (Sampling)
Chain 4 Iteration: 210 / 450 [46%] (Sampling)
Chain 4 Iteration: 220 / 450 [48%] (Sampling)
Chain 4 Iteration: 230 / 450 [51%] (Sampling)
Chain 4 Iteration: 240 / 450 [53%] (Sampling)
Chain 4 Iteration: 250 / 450 [55%] (Sampling)
Chain 1 Iteration: 260 / 450 [57%] (Sampling)
Chain 1 Iteration: 270 / 450 [60%] (Sampling)
Chain 1 Iteration: 280 / 450 [62%] (Sampling)
Chain 1 Iteration: 290 / 450 [64%] (Sampling)
Chain 1 Iteration: 300 / 450 [66%] (Sampling)
Chain 1 Iteration: 310 / 450 [68%] (Sampling)
Chain 1 Iteration: 320 / 450 [71%] (Sampling)
Chain 2 Iteration: 290 / 450 [64%] (Sampling)
Chain 2 Iteration: 300 / 450 [66%] (Sampling)
Chain 2 Iteration: 310 / 450 [68%] (Sampling)
Chain 2 Iteration: 320 / 450 [71%] (Sampling)
Chain 2 Iteration: 330 / 450 [73%] (Sampling)
Chain 2 Iteration: 340 / 450 [75%] (Sampling)
Chain 2 Iteration: 350 / 450 [77%] (Sampling)
Chain 3 Iteration: 260 / 450 [57%] (Sampling)
Chain 3 Iteration: 270 / 450 [60%] (Sampling)

Chain 3 Iteration: 280 / 450 [62%] (Sampling)
Chain 3 Iteration: 290 / 450 [64%] (Sampling)
Chain 3 Iteration: 300 / 450 [66%] (Sampling)
Chain 3 Iteration: 310 / 450 [68%] (Sampling)
Chain 3 Iteration: 320 / 450 [71%] (Sampling)
Chain 3 Iteration: 330 / 450 [73%] (Sampling)
Chain 4 Iteration: 260 / 450 [57%] (Sampling)
Chain 4 Iteration: 270 / 450 [60%] (Sampling)
Chain 4 Iteration: 280 / 450 [62%] (Sampling)
Chain 4 Iteration: 290 / 450 [64%] (Sampling)
Chain 4 Iteration: 300 / 450 [66%] (Sampling)
Chain 4 Iteration: 310 / 450 [68%] (Sampling)
Chain 4 Iteration: 320 / 450 [71%] (Sampling)
Chain 4 Iteration: 330 / 450 [73%] (Sampling)
Chain 1 Iteration: 330 / 450 [73%] (Sampling)
Chain 1 Iteration: 340 / 450 [75%] (Sampling)
Chain 1 Iteration: 350 / 450 [77%] (Sampling)
Chain 1 Iteration: 360 / 450 [80%] (Sampling)
Chain 1 Iteration: 370 / 450 [82%] (Sampling)
Chain 1 Iteration: 380 / 450 [84%] (Sampling)
Chain 1 Iteration: 390 / 450 [86%] (Sampling)
Chain 1 Iteration: 400 / 450 [88%] (Sampling)
Chain 2 Iteration: 360 / 450 [80%] (Sampling)
Chain 2 Iteration: 370 / 450 [82%] (Sampling)
Chain 2 Iteration: 380 / 450 [84%] (Sampling)
Chain 2 Iteration: 390 / 450 [86%] (Sampling)
Chain 2 Iteration: 400 / 450 [88%] (Sampling)
Chain 2 Iteration: 410 / 450 [91%] (Sampling)
Chain 2 Iteration: 420 / 450 [93%] (Sampling)
Chain 3 Iteration: 340 / 450 [75%] (Sampling)
Chain 3 Iteration: 350 / 450 [77%] (Sampling)
Chain 3 Iteration: 360 / 450 [80%] (Sampling)
Chain 3 Iteration: 370 / 450 [82%] (Sampling)
Chain 3 Iteration: 380 / 450 [84%] (Sampling)
Chain 3 Iteration: 390 / 450 [86%] (Sampling)
Chain 3 Iteration: 400 / 450 [88%] (Sampling)
Chain 4 Iteration: 340 / 450 [75%] (Sampling)
Chain 4 Iteration: 350 / 450 [77%] (Sampling)
Chain 4 Iteration: 360 / 450 [80%] (Sampling)
Chain 4 Iteration: 370 / 450 [82%] (Sampling)
Chain 4 Iteration: 380 / 450 [84%] (Sampling)
Chain 4 Iteration: 390 / 450 [86%] (Sampling)
Chain 4 Iteration: 400 / 450 [88%] (Sampling)

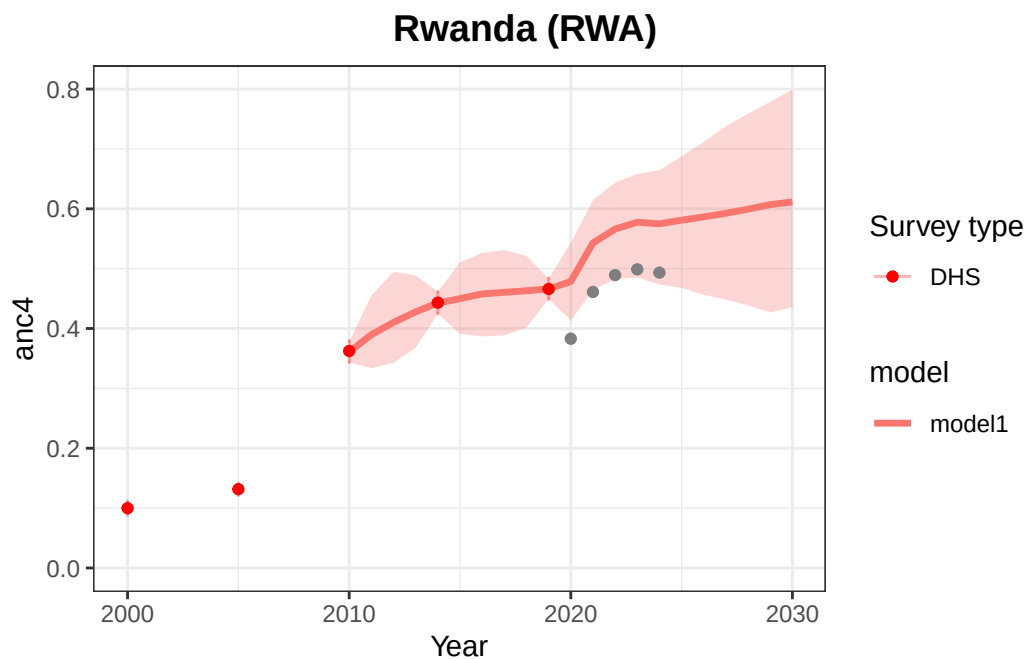
Chain 1 Iteration: 410 / 450 [91%] (Sampling)
Chain 1 Iteration: 420 / 450 [93%] (Sampling)
Chain 1 Iteration: 430 / 450 [95%] (Sampling)
Chain 1 Iteration: 440 / 450 [97%] (Sampling)
Chain 1 Iteration: 450 / 450 [100%] (Sampling)
Chain 2 Iteration: 430 / 450 [95%] (Sampling)
Chain 2 Iteration: 440 / 450 [97%] (Sampling)
Chain 2 Iteration: 450 / 450 [100%] (Sampling)
Chain 3 Iteration: 410 / 450 [91%] (Sampling)
Chain 3 Iteration: 420 / 450 [93%] (Sampling)
Chain 3 Iteration: 430 / 450 [95%] (Sampling)
Chain 3 Iteration: 440 / 450 [97%] (Sampling)
Chain 3 Iteration: 450 / 450 [100%] (Sampling)
Chain 4 Iteration: 410 / 450 [91%] (Sampling)
Chain 4 Iteration: 420 / 450 [93%] (Sampling)
Chain 4 Iteration: 430 / 450 [95%] (Sampling)
Chain 4 Iteration: 440 / 450 [97%] (Sampling)
Chain 4 Iteration: 450 / 450 [100%] (Sampling)
Chain 1 finished in 1.0 seconds.
Chain 2 finished in 0.9 seconds.
Chain 3 finished in 1.0 seconds.
Chain 4 finished in 0.9 seconds.

All 4 chains finished successfully.
Mean chain execution time: 1.0 seconds.
Total execution time: 1.1 seconds.

Extracting posteriors...

Overview plot

```
countryplot <- plot_estimates_local_all(  
  results = fit,  
  save_plots = FALSE)[[1]]  
countryplot
```



Estimates

```
fit$posteriors$temporal %>%
  select(iso, year, `50%`, `2.5%`, `97.5%`)
```

```
# A tibble: 21 x 5
  iso   year `50%` `2.5%` `97.5%`
  <chr> <int> <dbl> <dbl> <dbl>
1 RWA   2010 0.361 0.343 0.377
2 RWA   2011 0.390 0.334 0.455
3 RWA   2012 0.410 0.342 0.495
4 RWA   2013 0.428 0.368 0.489
5 RWA   2014 0.443 0.425 0.461
6 RWA   2015 0.450 0.391 0.511
7 RWA   2016 0.458 0.387 0.527
8 RWA   2017 0.461 0.388 0.531
9 RWA   2018 0.463 0.402 0.521
10 RWA  2019 0.466 0.449 0.483
# i 11 more rows
```